

Luis Alberto Ramos

Senior Software Engineer | Backend & Distributed Systems | Full Stack & Mobile Developer

Colombia · GitHub: github.com/lrcliche · Portfolio: lrcliche.github.io

PROFESSIONAL SUMMARY

Senior Software Engineer with over 7 years of hands-on experience designing and delivering scalable backend services, microservices architectures, industrial IoT integrations, and cross-platform mobile applications. Proven track record in core industrial backend development with **Java**, high-throughput microservices in **Go** and **Node.js/NestJS**, full-stack development with **React/Next.js/Vue**, cross-platform mobile apps (**Flutter/Dart**, **React Native**, **Vue + Capacitor**), and database administration (**PostgreSQL**, **MongoDB**, **SQL Server**, **Redis**). Expert in event-driven architectures (RabbitMQ, MQTT, WebSockets, gRPC), production Linux server administration, and technical leadership.

TECHNICAL SKILLS

- **Languages:** Java, Go (Golang), Node.js, TypeScript, JavaScript, Python (*Personal Projects*), SQL, Dart
- **Backend & Architecture:** Microservices, Hexagonal Architecture (Ports & Adapters), REST APIs, gRPC, WebSockets, NestJS, Gin, DDD Principles, SOLID
- **Frontend & Mobile:** React, Next.js, Vue 3, Vite, Tailwind CSS, Flutter, React Native, Expo, Vue + Capacitor (Mobile POS)
- **Databases & Cloud Storage:** PostgreSQL (DBA & Query Tuning), MongoDB, Firebase, SQL Server, MySQL, Redis, Cloud Storage (S3/Blob)
- **Messaging & Systems:** RabbitMQ, MQTT, Async Queues, Custom POS Event Processors, TCP/IP, Modbus, Serial Communication
- **DevOps & Infrastructure:** Linux / VPS Management, Docker, Nginx, Git, GitHub Actions (CI/CD), SonarQube, Incident Troubleshooting

PROFESSIONAL EXPERIENCE

Senior Backend / Full Stack Engineer & DBA — Devitech S.A.S (2019 — Present | Enterprise & Industrial)

- Developed and maintained core industrial backend systems utilizing **Java** and high-concurrency microservices using **Go** and **TypeScript/Node.js**.
- Integrated fuel dispenser hardware, **Veeder-Root** ATG systems, edge controllers, and payment pinpads over Modbus, TCP/IP, and RS-232/RS-485 serial protocols.
- Architected async event processing and messaging queues with **RabbitMQ**, **MQTT**, and **gRPC**, building custom resilient event handlers for POS operations.
- Administered and optimized **PostgreSQL** and **MongoDB** databases, executing query tuning, indexing, and high-volume data integration pipelines (API → MongoDB → PostgreSQL).
- Managed Linux VPS infrastructure, manual production deployments, **SonarQube** code quality audits, and live incident troubleshooting on production environments.

Lead Architect & Developer (SaaS Electronic Invoicing) — Factra (2021 — Present | Personal Product)

- Designed and implemented a microservices-based SaaS electronic invoicing engine with **Go**, **Node.js**, and **NestJS**, managing multi-tenant authentication via API Gateways.
- Engineered asynchronous queue consumers for high-volume tax document generation, cryptographic signing (XAdES/XMLDSig), and fault-tolerant submission to tax authority endpoints.
- Integrated payment gateway processors using **Go** and **Wompi** API webhooks with strict idempotency controls to eliminate duplicate transactions.
- Developed the web administration platform in **React** and **TypeScript** for inventory, sales transactions, client management, and real-time financial reporting.

FEATURED PROJECTS & PRODUCTS

- **Factra Mobile POS:** Cross-platform handheld POS app (*Vue 3, Vite, Capacitor, TypeScript*) with offline-first **SQLite** storage and automatic cloud synchronization.

- **EMAUS POS:** Full-stack multi-branch POS platform (*Next.js, React, NestJS, PostgreSQL*) featuring shift control, terminal ops, and inventory locking.
- **Go Hexagonal Architecture Template:** Open-source Go starter boilerplate (*Go, Gin, Docker*) enforcing Ports & Adapters architecture. [\[Repo Link\]](#)
- **Urban Mobility & Taxi Application:** Personal mobile project (*Flutter, React Native, Expo, WebSockets*) for real-time driver dispatch and geospatial location streaming.

EDUCATION & LANGUAGES

- **Software Engineering Background:** Continuous professional development, autodidact mastery, and 7+ years of verifiable production software delivery.
- **Languages:** Spanish (Native), English (Professional Working Proficiency for technical documentation and international teams).